

Australia Rain Prediction

From RAW-Data to Production - MLOps

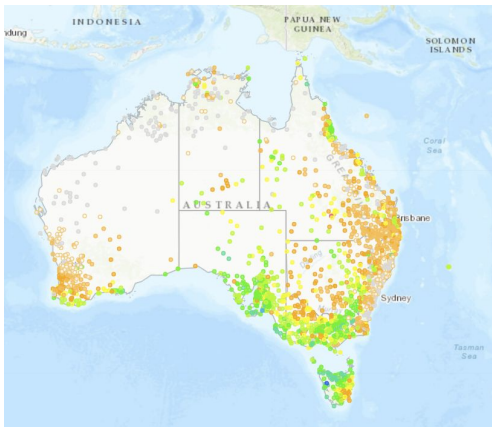
Binary Rain Classifier · Australian Weather Stations · Full MLOps Lifecycle

Question & Goal

Slide 2 / 13 · 2 min

Business Question

Will it rain tomorrow at a given Australian weather station? Useful for agriculture, logistics, and event planning.



Why It Matters

Binary forecast enabling timely operational decisions. ~22% of Melbourne days exceed 1 mm rainfall threshold.

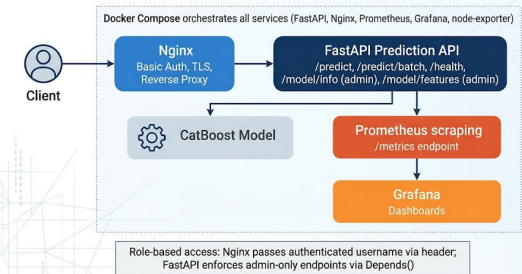
Melbourne (Olympic Park), Victoria
June 2026 Daily Weather Observations

Date	Day	Temps		Rain	Evap	Sun	Max wind gust			9 am			3 pm							
		Dir	Spd				Time	Temp	RH	Cld	Dir	Spd	MSLP	Temp	RH	Cld	Dir	Spd	MSLP	
		°C	°C	mm	mm	hours	km/h	local	°C	%	g/h	km/h	hPa	°C	%	g/h	km/h	hPa		
1	Mo	8.3	13.6	0			N	56	23.03	9.6	78	N	13	1012.4	11.4	85	N	15	1006.3	
2	Tu	9.6	17.6	7.4			N	48	00.15	13.6	77	N	19	994.0	16.4	72	NNW	17	991.9	
3	We	12.8	16.4	0.2			NNW	46	09.40	13.7	68	NNW	17	996.9	15.7	54	NNW	19	996.9	
4	Th	10.8	14.2	0			NNW	43	01.35	11.2	69	N	15	998.4	13.5	66	SSW	9	999.2	
5	Fr	9.8	15.3	18.6			SSW	30	12.51	11.0	76	W	9	1017.5	14.9	54	W	10	1017.5	
6	Sa	10.9	16.1	0			WSW	22	14.00	13.7	83	NNW	9	1025.6	15.9	78	ENE	5	1206.2	
7	Su	10.4	18.4	1.2			N	19	13.01	12.4	90	NE	7	1030.5	16.7	73			Calm	1022.9
8	Mo	8.5	16.6	0			NNW	28	15.35	10.4	83	NNE	9	1031.7	15.6	65	NNW	15	1028.5	
9	Tu	6.7	19.4	0			NNW	39	16.09	11.8	76	N	9	1024.9	17.6	61	N	19	1020.8	
10	We	11.8	18.5	0.4			N	52	01.38	15.9	84	N	17	1019.6	18.1	66	NNW	13	1020.6	
11	Th	8.2	17.8	1.4			N	15	10.01	9.9	100	NE	7	1032.1	16.5	63	ESE	4	1026.6	
12	Fr	9.9	19.1	0			N	54	11.16	15.6	81	N	9	1030.1	18.6	65	N	10	1029.8	
13	Sa	15.5	16.7	0			N	46	00.30	15.8	74	N	26	1022.1	16.0	83	N	13	1020.1	
14	Su	7.6	14.1	10.6			SSW	17	13.18	9.3	100		Calm	1030.6	13.5	69	SSW	7	1030.7	
15	Mo	5.1	17.6	0			NNW	43	13.49	7.2	94	ENE	6	1031.1	17.1	74	N	17	1020.5	
16	Tu	7.2	18.2	0			N	44	03.20	16.4	81	N	17	1024.9	17.3	76	N	15	1023.2	
17	We	14.5	20.0	0.8			N	37	11.30	16.3	82	NNE	7	1023.6	19.0	72	N	17	1020.5	
18	Th	16.2	18.3	0.4			NNW	46	08.14	18.0	70	N	20	1011.2	16.9	77	N	28	1006.4	
19	Fr	11.4	14.9	1.0			N	44	10.07	13.2	72	N	15	1005.2	14.7	70	N	20	1005.2	
20	Sa	11.5	14.4	5.2			SSW	33	10.30	12.3	96	SW	11	1019.3	13.5	70	SSW	15	1022.3	
21	Su	9.8	15.1	1.4			S	22	14.29	11.2	91	SE	6	1030.1	14.1	72	SSW	13	1022.9	
22	Mo	7.0	15.1	0			SSW	17	14.29	8.1	97	NE	7	1029.6	12.9	79	SSW	7	1029.7	
23	Tu	6.1		0							80	93	NE	7	1028.6	14.8	67	N	9	1027.3
Statistics for the first 23 days of June 2026																				
Mean 10.0 16.7																				
Lowest 5.1 13.6																				
Highest 16.2 20.0																				
Total 48.6																				
N 56 18.0 100 N 26 1032.1 19.0 86 N 28 1030.7																				
DC:JOW3033 202606 Prepared at 05:37 UTC on Tuesday 23 June 2026																				

Project Goal

Not only maximize predictive performance — demonstrate a reproducible, versioned, monitored MLOps lifecycle end-to-end.

MLOps Architecture Overview



⚡ ML + Ops — not just a model that works once, but a pipeline that can be retrained, tracked, deployed, and monitored repeatedly.

Dataset Overview

Slide 3 / 13 · 3 min

145,461

Raw Rows

2008 – 2017

23

Features

weather metrics

49

Stations

across Australia

~22%

Rain Days

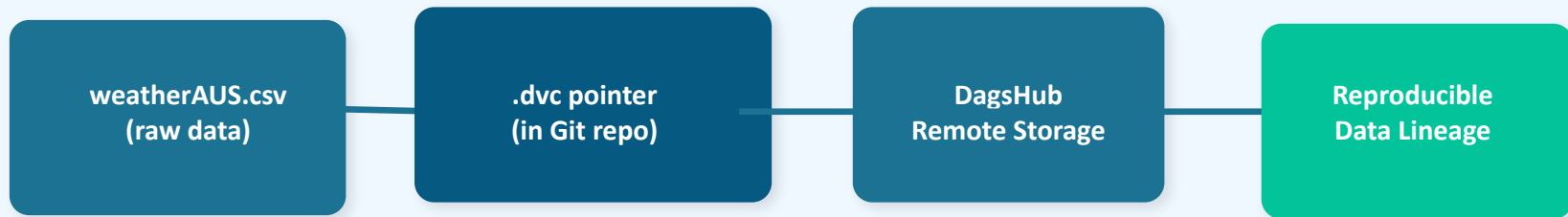
class imbalance

Key Features

Feature Group	Examples	Count
Numeric	Humidity, Pressure, Temperature, Wind, Rainfall...	16
Categorical	Location, WindGustDir, WindDir9am/3pm, Season	6
Engineered	Season (from Date → Summer/Autumn/Winter/Spring)	1
Target	RainToday (Yes/No) — renamed from RainTomorrow	1

Data Versioning — DVC + DagsHub

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Why DVC?

- Raw data & model artifacts NOT stored in Git — only lightweight .dvc pointer files
- Content-addressed blobs on DagsHub: every data version is fully reproducible
- Prevents repository bloat while maintaining complete data lineage
- Real lesson learned: DVC remote had gaps (blobs never pushed) → recovered from original source, re-ran dvc push to restore consistency

⚠ Lesson: DVC isn't just 'git for data' — remote storage gaps are a real failure mode. Always verify dvc push in CI.

Feature Engineering & Preprocessing

Slide 5 / 13 · 3 min

Data Pipeline Steps

- 1 Load** 145,460 rows × 23 cols from BOM/Kaggle CSV
- 2 dropna()** → 56,420 rows (removes all rows with ANY missing value)
- 3 Location Filter** Melbourne, MelbourneAirport, Watsonia → 7,557 rows
- 4 Rename Target** RainToday → RainYesterday | RainTomorrow → RainToday
- 5 Season Feature** Date.month mapped to Summer/Autumn/Winter/Spring
- 6 Drop Date** Season captures temporal signal; raw date dropped

Sklearn Preprocessing Pipeline

Numeric (16 features)

StandardScaler → zero mean, unit variance

Categorical (6 features)

OneHotEncoder (handle_unknown='ignore')

ColumnTransformer → unified feature matrix



No Leakage: imputation/scaling statistics computed on X_train only — applied to X_test. Stratify=y preserves 76/24 class ratio in splits.

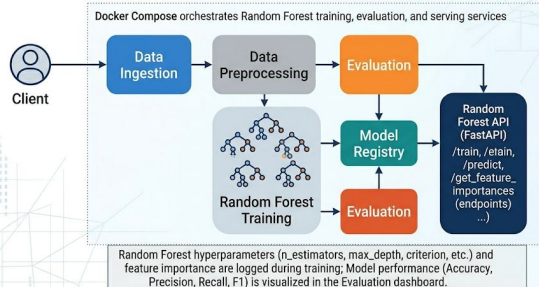
Modeling Pipeline

Slide 6 / 13 · 3 min

Random Forest (Primary)

- `n_estimators`: [50, 100]
- `max_depth`: [None, 10, 20]
- `min_samples_split`: [2, 5]
- → 12 combos × 5 folds = 60 fits
- `StratifiedKfold(n_splits=5, shuffle=True)`
- Scoring: accuracy

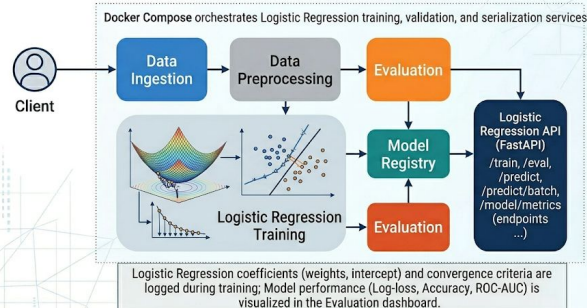
MLOps Architecture Overview



Logistic Regression (Baseline)

- `solver`: ['liblinear']
- `penalty`: ['l1', 'l2']
- `class_weight`: [None, 'balanced']
- → 4 combos × 5 folds = 20 fits
- Same StratifiedKFold strategy
- Note: `penalty` deprecated in `sklearn` ≥ 1.8

MLOps Architecture Overview - Logistic Regression



Chronological discipline: split respects time-series nature of weather data → no future pattern leakage. Full pipeline: Preprocessor → Classifier → GridSearchCV.

Model Results

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~85%

RF Accuracy

Random Forest

83%

LR Accuracy

Logistic Regression

76%

Baseline

Always predict No Rain

51%

LR Recall

Rain class (Yes)

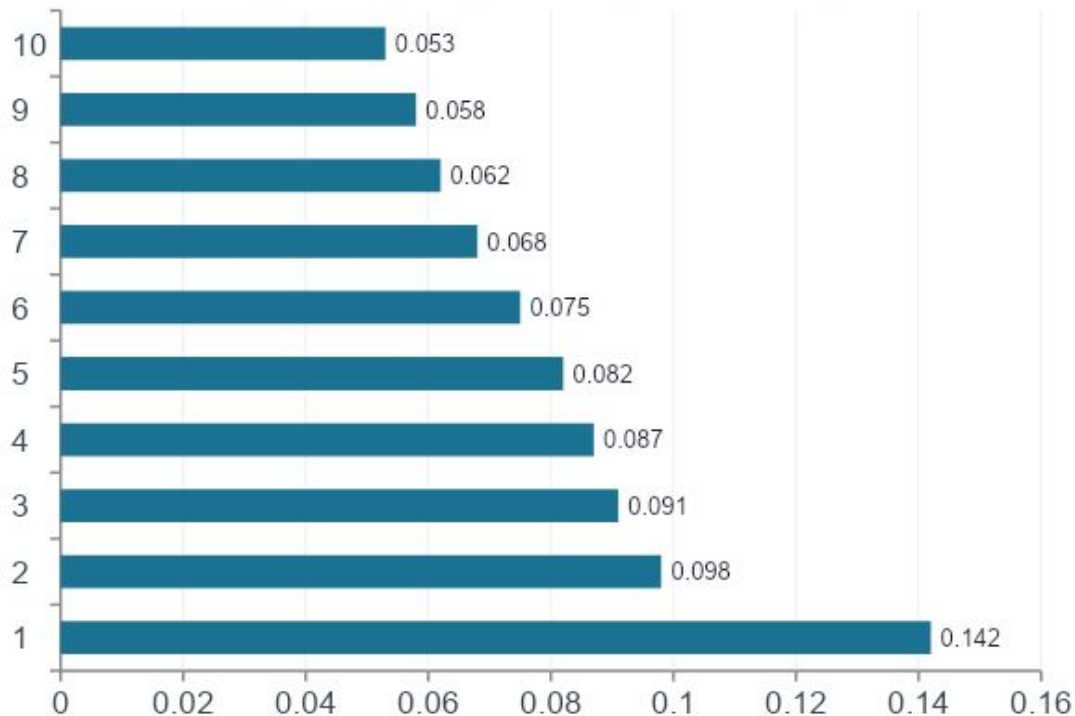
Detailed Comparison

Metric	Random Forest	Logistic Regression	Advantage
Accuracy	~85%	83%	RF
Precision (Rain)	Higher	0.68	RF
Recall (Rain)	Higher	0.51	RF
F1 (Rain)	Higher	0.58	RF
Interpretability	Medium (feat. importance)	High (coefficients)	LR
Training Time	~0.4–0.9s/fold	~0.1–0.2s/fold	LR

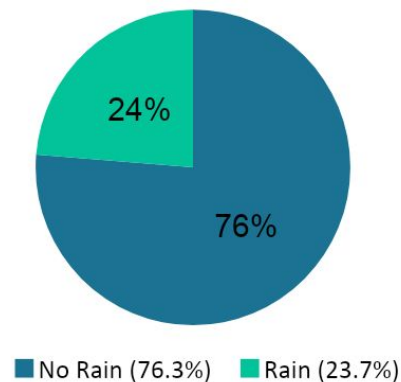
Model Results — Visual Analysis

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Top 10 Feature Importances (Random Forest)



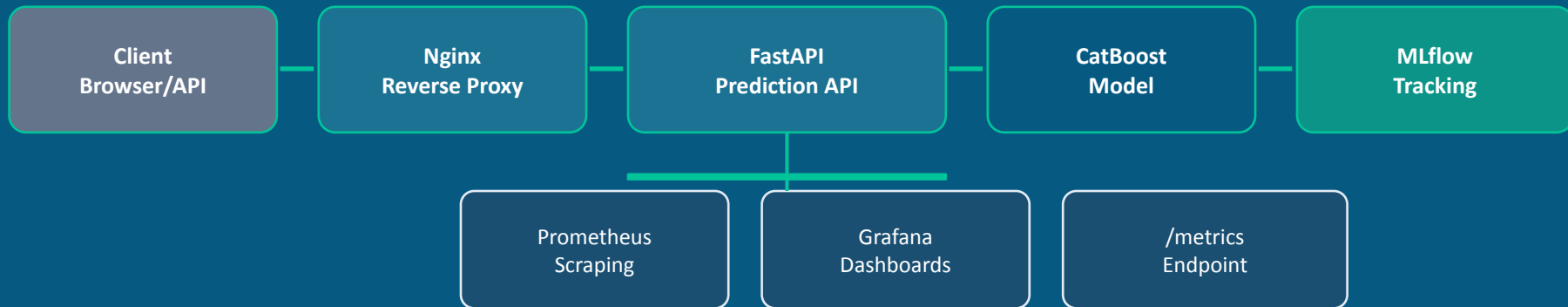
Class Balance (n=7,557)



Key Insight

- Humidity3pm is the #1 predictor
- Followed by pressure measurements
- Imbalanced classes → always use stratify=y

MLOps Architecture Overview



API Endpoints

POST	/predict	→ RainToday: Yes/No + confidence
GET	/health	→ service status
GET	/metrics	→ Prometheus metrics
GET	/model/info	→ active model version (admin)
GET	/model/features	→ feature schema (admin)

Security & Access Control

- Nginx: Basic Auth + TLS / SSL termination
- Role-based access: admin vs. user via Depends()
- Auth username passed via header to FastAPI
- Docker Compose orchestrates all services
- ⚠ Credentials externalized to env vars (roadmap)

Experiment Tracking — MLflow + DagsHub

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What Gets Logged Per Run



Hyperparameters

n_estimators, max_depth, min_samples_split (RF) / solver, penalty, class_weight (LR)



Evaluation Metrics

accuracy, precision, recall, F1 (per class + weighted avg)



Test Period Dates

Train/test date ranges for reproducibility



Model Artifact

Trained sklearn pipeline (CatBoost / RF / LR) logged as MLflow artifact



DVC Pointer

Link to versioned data snapshot used in training run

MLflow Integration Snippet

```
import mlflow

with mlflow.start_run():

    mlflow.log_params(grid_search.best_params_)
    mlflow.log_metric('accuracy',
                      best_score)
    mlflow.sklearn.log_model(

        grid_search.best_estimator_,

        'rain_classifier'

    )
```

The screenshot shows the MLflow web interface for experiment 'final_hybrid_catboost'. The 'Overview' tab is selected, displaying metadata: Created at 06/23/2026, 03:17:31 PM; Created by gnembac; Experiment ID 0; Status Finished; Run ID 63c7f3a2cb544a6f94a4d416f5c0e941; Duration 2.5min; Source train_winner.py. The 'About this run' section is also visible.

The screenshot shows the DagsHub repository page for 'Inesgas / rain_prediction_MLops'. It displays a file tree with folders for .dvc, .github, data, and docker. The 'data' folder is highlighted, showing its commit history and a 'Create dataset' button. The right sidebar shows repository metadata like collaborators and topics.

Monitoring — Prometheus + Grafana

Slide 11 / 13 · 5 min

Infrastructure Metrics (Auto-instrumented)

✓ Request latency (histogram)

✓ Request counts by endpoint

✓ HTTP status code distribution

✓ Error rate

✓ CPU / Memory (node-exporter)

Custom ML / Business Metrics

`rain_predictions_total`

predictions by location + outcome

`rain_prediction_confidence`

confidence score histogram

`rain_input_humidity/rainfall/pressure`

data drift signal per location

`rain_prediction_errors_total`

error counter by failure type

🎯 Drift Monitoring: Humidity3pm & Pressure9am gauges serve as early-warning system when incoming feature distributions shift from training data.

Limitations & Lessons Learned

Slide 12 / 13 · 3 min

DVC Remote Gaps

Missing blobs discovered during project — never pushed initially. Recovered from original source, re-ran dvc push to restore team consistency.

Hardcoded Credentials

User/role credentials originally in API code. Identified as security risk — proposed fix: externalize to environment variables / secrets manager.

Unrecoverable Artifacts

3 baseline feature tables (~300 MB total) remain unreproducible — no script regenerates them from raw data. Documentation gap.

Coarse-grained Auth

Role-based access is admin vs. user only. No per-user rate limiting or audit logging yet. Acceptable for MVP, not for production.

Being upfront about real gaps is a strength in an MLOps presentation — it shows production-mindedness, not just 'it works on my machine'.

Roadmap & Closing

Next Steps

- 1 Externalize all credentials fully (env vars / secrets manager)
- 2 Automate dvc push + data integrity checks in CI pipeline
- 3 Add MLflow model registry stage transitions: staging → production
- 4 Automated retraining trigger on data drift signal from Grafana alerts
- 5 Extend per-user rate limiting and audit logging in FastAPI

"This project shows the full lifecycle — not just a model that works once, but a pipeline that can be retrained, tracked, deployed, and monitored repeatedly."